



# PLANET 4

## K2-18 B

**Description:** A sub-Neptune wrapped in a thick hydrogen atmosphere. Whether it hides a deep ocean underneath is still debated.

**Host star:** K2-18, a cool red dwarf about a third of the mass of the Sun.

**System:** One of 2 known planets orbiting K2-18, and the outer of the two.

**Location:** Constellation Leo, about 124 light-years from Earth. Much too faint for the naked eye (magnitude 13.5).

**Discovery:** In 2015 by NASA's K2 mission, the repurposed Kepler space telescope, led by Benjamin Montet et al.

## Datasheet

| EXOPLANET DATA                                                                      |                  |                                  |
|-------------------------------------------------------------------------------------|------------------|----------------------------------|
|    | TEMPERATURE      | -8 °C                            |
|   | PRESSURE         | Unknown                          |
|  | ATMOSPHERE       | H <sub>2</sub> , CH <sub>4</sub> |
|  | GRAVITY          | 1.59 g                           |
|  | PLANET RADIUS    |                                  |
|  | ORBITAL PERIOD   |                                  |
|  | SEMI-MAJOR AXIS  |                                  |
|  | TRANSIT DEPTH    |                                  |
|  | TRANSIT DURATION |                                  |

| HOST STAR DATA                                                                      |                    |                      |
|-------------------------------------------------------------------------------------|--------------------|----------------------|
|    | NAME               | K2-18                |
|   | SPECTRAL TYPE      | M2.5V                |
|  | TEMPERATURE        | 3457 K               |
|  | DISTANCE           | 124 ly               |
|  | MASS               | 0.359 M <sub>☉</sub> |
|  | STAR RADIUS        | 0.411 R <sub>☉</sub> |
|  | AGE                | Unknown              |
|  | APPARENT MAGNITUDE | 13.48                |
|  | NUMBER OF PLANETS  | 2                    |

### SURVIVAL CHALLENGE

No sunlight reaches the deep ocean floor beneath a thick hydrogen atmosphere

### NOTES



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| NOTES |
|-------|
|       |

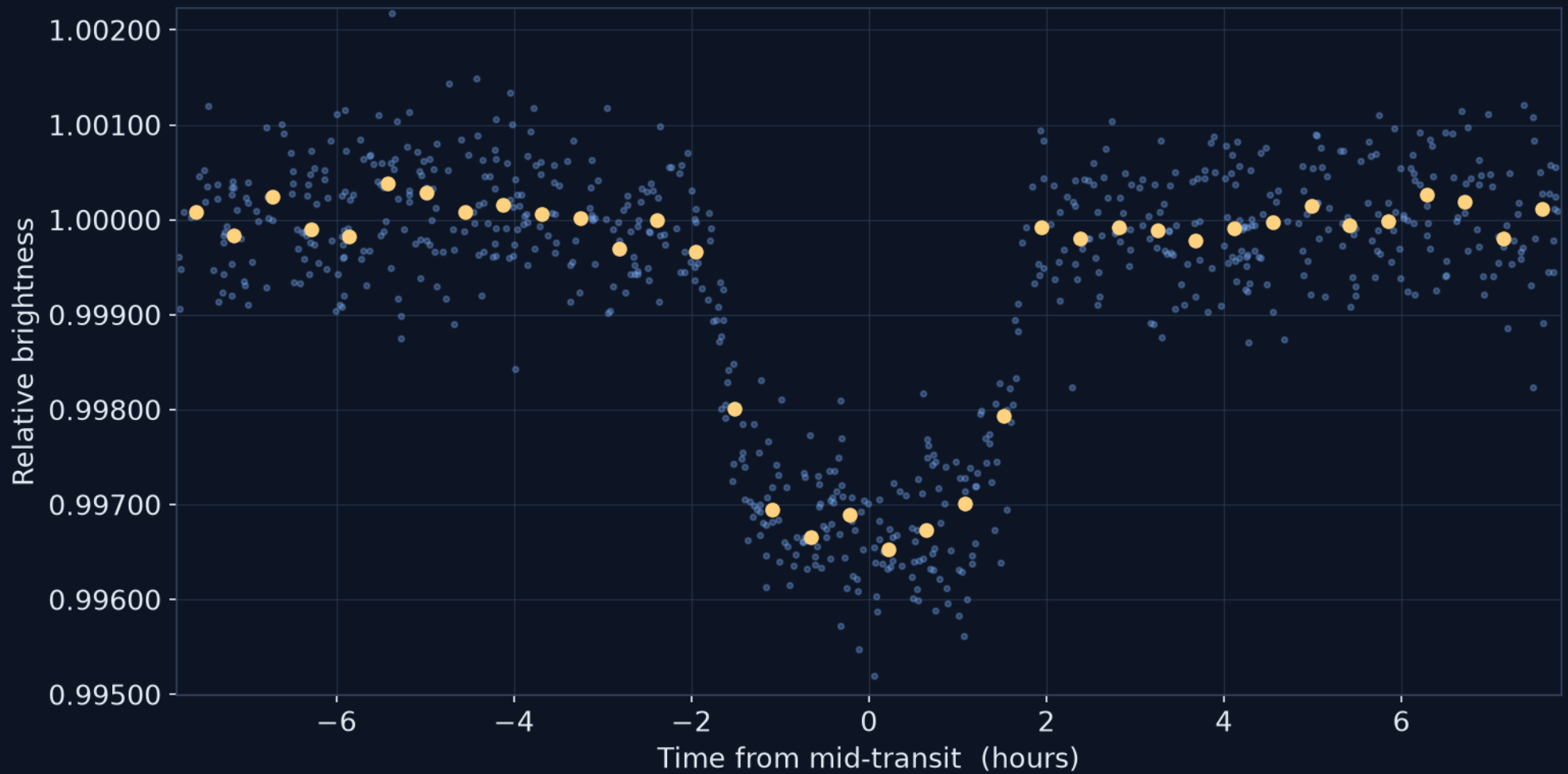
| PHYSICS                                                                                         |  |
|-------------------------------------------------------------------------------------------------|--|
|  PLANET RADIUS |  |
|  AI-ASSISTANCE |  |
|  REFLECTIONS   |  |

| MATHEMATICS                                                                                      |  |                                                                                                     |  |
|--------------------------------------------------------------------------------------------------|--|-----------------------------------------------------------------------------------------------------|--|
|  ORBITAL PERIOD |  |  SEMI-MAJOR AXIS |  |
|  AI-ASSISTANCE  |  |                                                                                                     |  |
|  REFLECTIONS    |  |                                                                                                     |  |

| CHEMISTRY                                                                                        |  |
|--------------------------------------------------------------------------------------------------|--|
|  SAMPLE DATA    |  |
|  AI-ASSISTANCE |  |
|  REFLECTIONS  |  |

| BIOLOGY                                                                                           |  |
|---------------------------------------------------------------------------------------------------|--|
|  ORGANISM      |  |
|  AI-ASSISTANCE |  |
|  REFLECTIONS   |  |

## K2-18 b: single transit



Blue points are simulated brightness measurements (many orbits phase-folded together, like real data); the gold points are binned averages.

K2-18 b: light curve over several orbits

